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## chapter thirteen

# SIMPLE LINEAR REGRESSION

*But of all these principles,  
least squares is the most simple:  
by the others we would be  
led into the most complicated calculations.*

– K.F. Gauss (1809)

### INTRODUCTION

All of our work thus far has involved a single variable. However, many of the problems that we work with in statistics involve two variables. The sets of data for these problems contain ordered pairs. The first value of the ordered pair is called the **independent variable**, and the second value of the ordered pair is called the **dependent variable**. We often refer to our work with two variables as bivariate data analysis. We divide the analysis into two areas, **regression analysis** and **correlation analysis**.

One of the first things we do with bivariate data is plot the ordered pairs on a rectangular coordinate system. In many cases the graph looks like the points are clustered around a straight line. We then say that there appears to be a linear relationship between the two variables. Our first problem is to find the equation of the line that best

explains the linear relationship. We use the familiar **slope-intercept form**,  $y = a + bx$ , of an equation for a line from algebra. The equations we use for  $a$  and  $b$  involve the  $\Sigma$  notation that we worked with in chapter 3. In particular, we will use the sums,  $\Sigma x$ ,  $\Sigma y$ ,  $\Sigma x^2$ ,  $\Sigma y^2$ , and  $\Sigma xy$ .

In general, it is impossible to find a line that all the points will fall on. So we focus on the distances (**errors**) between the points and the line we draw. We want the line that minimizes the squares of the distances using the **least squares method**. This line is called the **line of best fit** or the **regression line**. We are then able to use the equation of this line to predict the approximate value of one variable when we know the value of the other variable. The approximate value we get from the regression line is a point estimate. However, we can also create an interval estimate for the value using a confidence interval.

If few of the points we plot are on the line, we say that the linear relationship is weak with a value of  $b$  close to 0. We may then want to test a hypothesis that there is no linear relationship between the two variables. This would be the same as saying that the true value,  $B$ , of the slope of the **population regression line** is 0. We can again use a  $t$  test to test the null hypothesis,  $H_0: B = 0$  against either a one-tailed or two-tailed alternate hypothesis.

If most of the points we plot are on the line or close to the line, we argue that there is a strong linear relationship between the two variables. We then feel good about using the equation for the regression line to predict values for one of the variables. We use what is called the linear **coefficient of determination** to measure the strength of the linear relationship. The coefficient of determination is the fraction of **explained variation** to the **total variation** in  $y$ .

The **correlation coefficient**,  $r$ , is the square root of the coefficient of determination. The correlation coefficient is a value between 0 and 1 for a line with a positive slope, and between -1 and 0 for a line with a negative slope. When all the points fall on the line,  $r = 1$  or  $r = -1$ .

## CHAPTER 13 HOT SPOTS

1. **How To Write The Equation Of A Line In Slope-Intercept Form.**  
Starts on **page 13–4**. Problems on **pages 13–4, 13–5, 13–6**.  
Mann 13.1.2, 13.2
2. **Estimating The Slope And Intercept Of The Regression Line.**  
Starts on **page 13–6**. Problems on **pages 13–9, 13–10**.  
Mann 13.2.1
3. **The Effect Of Errors On The Regression Line And The Correlation Coefficient.**  
Starts on **page 13–11**. Mann 13.2.2
4. **Using A  $t$  Test Statistic For A Hypothesis Test Of  $H_o: B = 0$  Versus  $H_a: B \neq 0$ .**  
Starts on **page 13–13**. Problems on **pages 13–14, 13–15**.  
Mann 13.5.3
5. **The Danger Of Using The Regression Line To Predict Values Of  $y$  For  $x$  Outside The Interval Used To Obtain The Line.**  
Starts on **page 13–15**. Mann 13.2.3, 13.2.5, 13.9

If you find other HOT SPOTS, write them down and use them as a focus of your discussion in the study group. Or you can use the HOT SPOTS as the topic for a help session with your professor.



Mann  
Sections 13.1.2  
And 13.2

### Hot Spot #1 – How To Write The Equation Of A Line In Slope-Intercept Form

In an algebra class we learn that the equation of a line can be written in the form  $y = a + bx$ , where  $b$  is the slope of the line and  $a$  is the  $y$  intercept. This form is known as the slope-intercept form for the equation of a line. Once we know the slope-intercept form of the equation, we can use the  $y$  intercept and slope of the line to graph the line.

The slope-intercept form we use in regression analysis is very much like going backwards from what we do in an algebra class. In algebra, we change an equation to the slope-intercept form to help us graph the line. In statistics, we have a set of ordered pairs and our problem is to find the equation of the line that best fits the bivariate data. We can use a scattergram to plot the data and draw what we believe to be a good estimate of the line of best fit. We can then use the line to find approximate values for the slope and intercept. Once we have these values, we can write the equation of the line given by our estimates.

In regression analysis, we only have a set of ordered pairs, and our problem is to find the equation of the line that best fits the bivariate data. The next **Hot Spot** shows how we can “eyeball” our best estimate for the line that fits the data. We can then use the line to find approximate values for the slope and intercept. Once we have these values, we can write the equation of the line given by our estimates.

The following example gives an illustration of what we do in an algebra class when we use the slope-intercept form of the equation to graph a line.

**Hot Spot #1 Sample Problem and Solution:** Given the equation  $2x + 3y = 4$ , use the slope-intercept form to graph the related line.

Solution: Step 1. We solve for  $y$  to get the slope-intercept form as follows:

$$y = -\frac{2}{3}x + \frac{4}{3}$$

Step 2. The coefficient of  $x$  is the slope  $b = -\frac{2}{3}$ . The constant is the  $y$  intercept  $a = \frac{4}{3}$ .

Step 3. To graph the line, we first locate the  $y$  intercept at  $(0, \frac{4}{3})$  and then use the slope to get a second point by starting at the  $y$  intercept and going 3 units to the right, followed by 2 units down. The graph in Figure 13-1 shows how it is done.

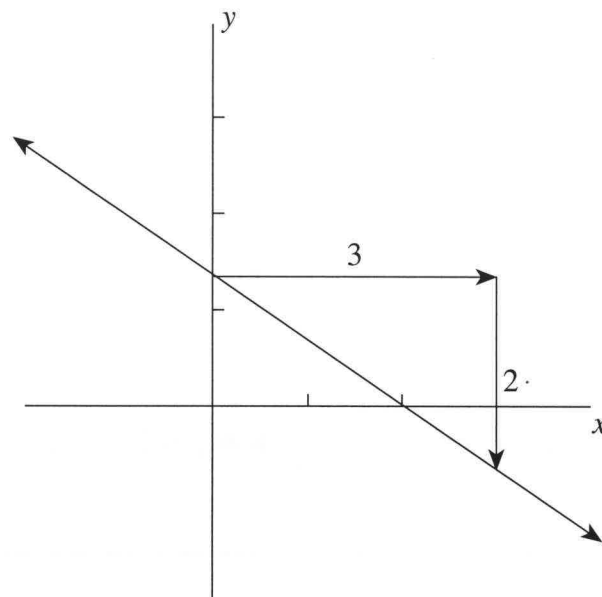
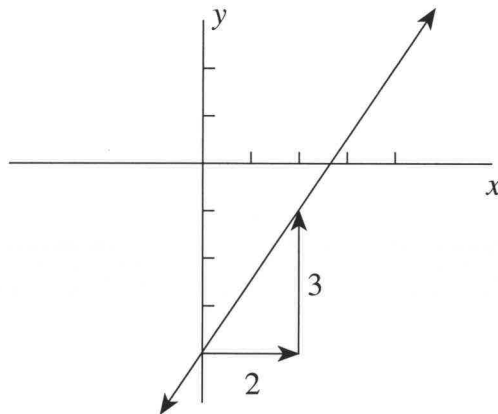


Figure 13-1:  
Function Graph  
 $y = -\frac{2}{3}x + \frac{4}{3}$

**Sample Problem and Answer:** Given the equation  $3x - 2y = 8$ , write the equation in slope-intercept and graph the line.

Answer:  $y = \frac{3}{2}x - 4$  and  $b = \frac{3}{2} = 1.5$  and  $a = -4$ . The graph is shown below, in Figure 13-2.

Figure 13-2:  
Using The  
Slope-Intercept  
Form To Graph  
 $3x-2y=8$



**Sample Problem and Answer:** Given the equation  $2x + 3y = 15$ , write the equation of the line in slope-intercept form and graph the line.

**Answer:**  $y = -\frac{2}{3}x + 5$  and  $b = -\frac{2}{3}$  and  $a = 5$ . The graph is shown in Figure 13-3.

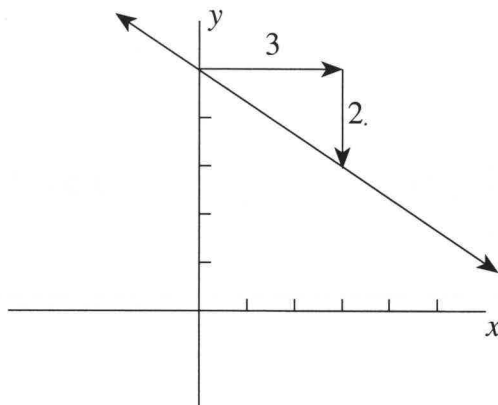


Figure 13-3:  
Using The  
Slope-Intercept  
Form To Graph  
 $2x+3y=15$

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Section 13.2.1

### Hot Spot #2 – Estimating The Slope And Intercept Of The Regression Line

When we have a set of bivariate data, we can plot the points and get what we call a scatter diagram. We then draw a line through the

data that to the best of our ability, contains as many points as possible, and minimizes the vertical distance from the points to the line for the points not on the line.

It may take some practice to consistently draw a line that is close to the actual line of least squares. Sometimes it helps to lightly draw several lines through the scatter diagram to get an idea of which one is best. Even then, there may be more than one line that seems acceptable. Just remember, we are only trying to get approximate values for the slope and intercept. If there are two equally reasonable lines, we go ahead and take one at random (perhaps flipping a coin if necessary to choose). The approximate values for the slope and  $y$  intercept that we get from the line will be our estimate for the values of the regression line we get using the least squares criterion.

The following scatter diagram (Figure 13-4) will be used as an example. Two lines have been drawn through the bivariate data set. The solid line will be used to estimate the values for the slope and  $y$  intercept.

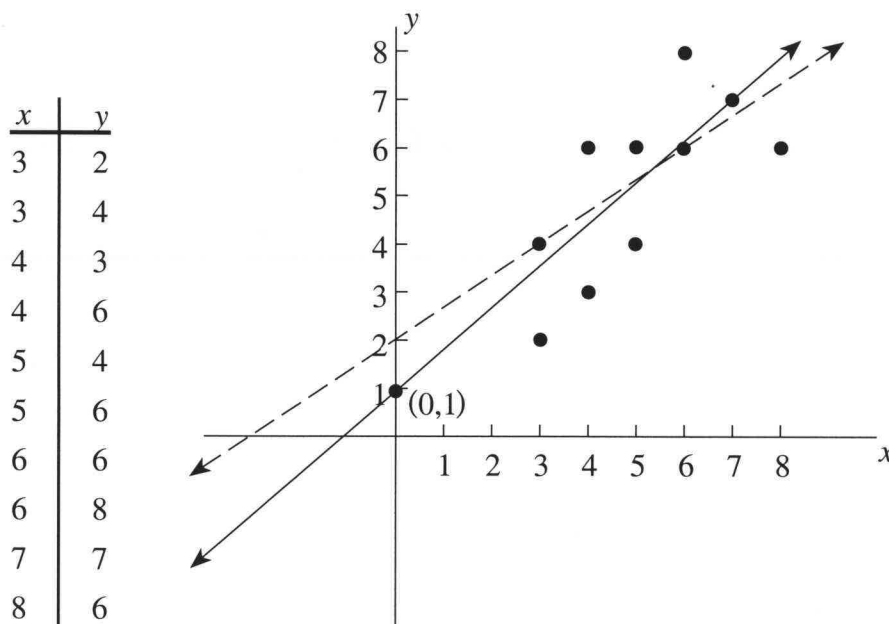
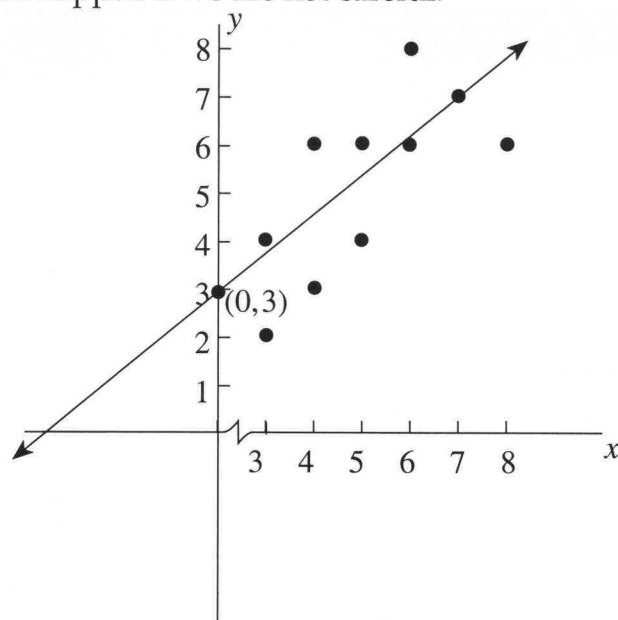


Figure 13-4:  
Drawing A  
Regression Line  
Through A  
Scatter Diagram

Figure 13-5:  
The Modified  
Form Of A  
Scatter Diagram  
With Incorrect  
Y Intercept



We need to extend the line so it cuts the vertical axis to find the value of the y intercept. In this example the line cuts the vertical axis at 1. We must be careful that both the  $x$  and  $y$  axis start at the origin without breaks. The modified form of the above graph (Figure 13-4) shows what can happen if we are not careful.

The  $y$  intercept appears to be positive 3 instead of the correct value of 1. The error is due to the break in the horizontal axis that was made to provide a more pleasing graph.

Returning to Figure 13-4, we find the slope by selecting two points on the line. It helps if the two points are actual ordered pairs from our bivariate data set that are on the line. We then use the slope formula from algebra to find the value of  $b$ .

$$b = \frac{y_2 - y_1}{x_2 - x_1}$$

We will use the ordered pairs (0,1) and (7,7) to find the slope.

$$b = \frac{7-1}{7-0} = \frac{6}{7} \doteq 0.857$$

We now have the approximate values for the slope and y intercept. The equation of our estimated line of least squares is

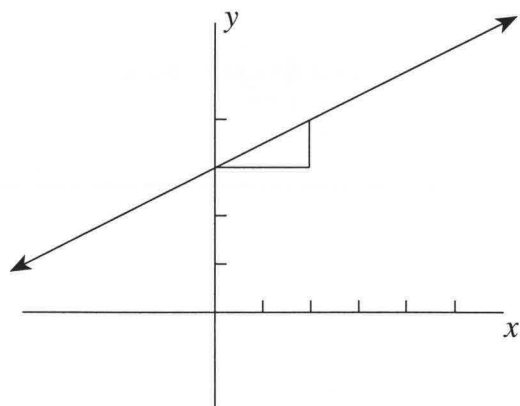
$$y = 0.857x + 1$$

As a comparison, we can get the equation of the line using the principle of least squares. We can use the equations in the text for the slope and intercept of the regression line, or we can use the stat package in the HP-21S. Either way, we get the following equation:

$$\hat{y} = 0.795x + 1.145$$

Our estimate was really quite good. We do not always do this well, but we can at least get an idea of whether or not the values we calculate using the equations for the slope and intercept are reasonable. Even if we use a calculator to get the equation of the regression line, it is important to have a visual sense of what the line should look like before we enter the data into the calculator.

**Hot Spot #2 Sample Problem and Solution:** Given the following graph of a line, find the slope and y intercept. Then write the equation of the line in the slope-intercept form  $y = a + bx$ .



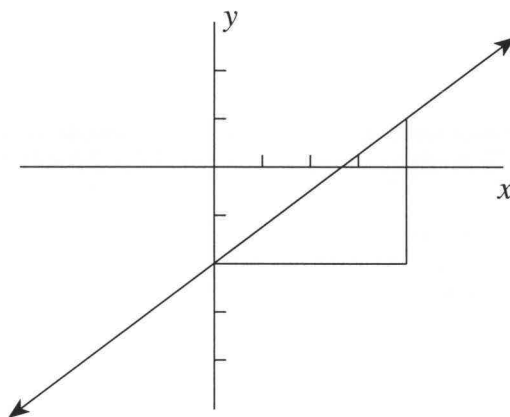
Solution: Step 1. The line crosses the y axis at (0,3).

Step 2. The slope of the line is  $\frac{1}{2}$ .

Step 3. The equation in slope-intercept form is  $y = \frac{1}{2}x + 3$ .

Figure 13-6:  
Finding Slope  
And Y Intercept  
From A Function  
Graph

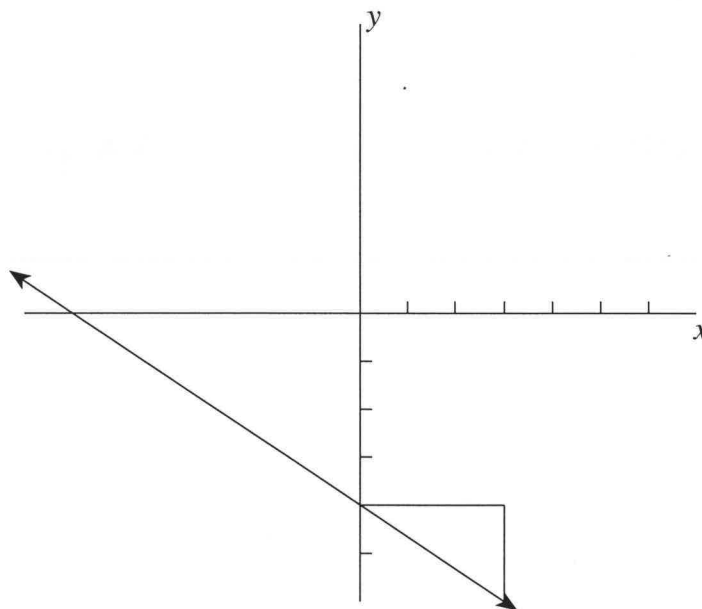
Figure 13-7:  
More Finding Of  
Slope And  
Y Intercept From  
A Function  
Graph



**Sample Problem and Answer:** Given the following graph of a line, find the slope and y intercept. Then write the equation of the line in the slope-intercept form  $y = a + bx$ .

Answer:  $b = \frac{3}{4}$ ,  $a = -2$ ,  $y = \frac{3}{4}x - 2$

Figure 13-8: Yet  
More Finding Of  
Slope And  
Y Intercept From  
A Function  
Graph



**Sample Problem and Answer:** Given the following graph of a line, find the slope and y intercept. Then write the equation of the line in the slope-intercept form  $y = a + bx$ .

Answer:  $b = -\frac{2}{3}$ ,  $a = -4$ ,  $y = -\frac{2}{3}x - 4$

### Hot Spot #3 – The Effect Of Errors On The Regression Line And The Correlation Coefficient

The formulas we use for finding the slope,  $b$ , and  $y$  intercept,  $a$ , of the line of least squares and the correlation coefficient,  $r$ , are very sensitive to errors in the summations we use. This is particularly true if the data values are small. The following set of bivariate data will be used to show what happens when errors are made:

| $x$ | $y$ | $x^2$ | $y^2$ | $xy$ |
|-----|-----|-------|-------|------|
| 1   | 1.2 | 1     | 1.44  | 1.2  |
| 1.5 | 0.8 | 2.25  | 0.64  | 1.2  |
| 2   | 0.6 | 4     | 0.36  | 1.2  |
| 3   | 0.4 | 9     | 0.16  | 1.2  |
| 4   | 0.3 | 16    | 0.09  | 1.2  |

The totals for each column are found as follows:

$$\sum x = 11.5 \quad \sum y = 3.3 \quad \sum x^2 = 32.25 \quad \sum y^2 = 2.69 \quad \sum xy = 6$$

When the above sums are substituted into the formulas for  $b$ ,  $a$ , and  $r$ , we get the following values:

$$b = -0.27413 \quad a = 1.2905 \quad r = -0.9227$$

We are now ready to see what happens if an error is made in one of the entries in the table. One of the most common mistakes is making an error in squaring one of the values in the table. The values of  $x$  and  $y$  in the above table are easy to square, but let's see what happens when we make an error in squaring  $y = 0.3$ . If  $(0.3)^2$  were written in error as 0.9, the only sum that would change is  $\sum y^2$ , which would increase to 3.5. In this case the values of  $b$  and  $a$  do not change because the formulas do not use  $\sum y^2$ . But there is a drastic change in the  $r$  value. The new value of  $r$  is -0.574205.

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Section  
13.2.2

Table 13-1:  
Table of Sample  
Bi-Variate Data

If we look at the value of  $r^2$  for the two different values of  $r$  we obtained, we can really see the effect of the error. The  $r^2$  of 0.8514 for the correct value of  $r$  tells us that approximately 85% of the variation in the  $y$  values is due to the regression line. When the error is made, the new  $r^2$  is approximately 33%, which is a reduction of more than 50% in explained variation.

Now let's look at errors that also affect values of  $b$  and  $a$ . If the error were made in a value of  $x^2$ , the error changes the value of the denominator in the formula for both  $b$  and  $r$  and thus changes the value of all three computations. We can illustrate this last point by looking at what happens if the square of 1.5 is written incorrectly as 22.5 rather than 2.25. The  $\Sigma x^2$  is increased by 20.25 and the effect on  $b$ ,  $a$ , and  $r$  is again drastic. The new values based on the error are as follows:

$$b = -0.0610 \qquad a = 0.8004 \qquad r = -0.4354$$

Of the two errors we examined this last one is the more serious because all three values are affected to such a degree that we might as well throw out the entire analysis. Again the point is that care must be taken in the calculations used in the formulas for  $b$ ,  $a$ , and  $r$ .

The obvious solution to this problem of errors is to use a calculator or computer to obtain the values for the regression line and the correlation coefficient. If we must do the calculations using the formulas, it is imperative that we avoid errors in the table of values we create to find  $b$ ,  $a$ , and  $r$ .

## Hot Spot #4 – Using A $t$ Test Statistic For A Hypothesis Test Of $H_o: B = 0$ Versus $H_a: B \neq 0$

The hypothesis test for the null hypothesis  $H_o: B = 0$  fits the template model introduced earlier in chapter 9 (page 9–5) where

|                                                                                                               |                                                                                                                                                                                                                       |
|---------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $z = \frac{\square - \mu_{\square}}{\sigma_{\square}}, \text{ where the } \square \text{ can be replaced by}$ | <ol style="list-style-type: none"> <li>1. <math>x</math></li> <li>2. <math>\bar{x}</math></li> <li>3. <math>\hat{p}</math></li> <li>4. <math>\bar{x}_2 - \bar{x}_1</math></li> <li>5. <math>\bar{d}</math></li> </ol> |
|---------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

We can now add the sample statistic  $b$  to our list:

6.  $b$

In the case of  $b$ , when  $B = 0$ , the sampling distribution for small samples with  $n < 30$  involves a  $t$  test with  $n - 2$  degrees of freedom. We lose the 2 degrees of freedom because we need to calculate both  $\bar{x}$  and  $\bar{y}$ .

Returning to our template, we need to replace  $z$  with  $t$ . When we place  $b$  in the box, the  $t$  test looks like the following:

$$t = \frac{b - \mu_b}{\sigma_b} \text{ where } \mu_b = B \text{ and } \sigma_b = \frac{\sigma_e}{\sqrt{SS_{xx}}} \text{ and } s_b = \frac{s_e}{\sqrt{SS_{xx}}}$$

The final form of our test statistic is:

$$t = \frac{b - B}{s_b} \text{ with } df = n - 2$$



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Section 13.5.3

Template For  $z$   
Test Statistic

Returning to the hypothesis test of  $B = 0$ , once we have a value of  $b$  and  $s_b$  the  $t$  test is easy to do. We follow the same five steps we have been using for hypothesis testing in the last three chapters.

- Step 1. State the hypotheses.
- Step 2. Select the distribution to use.
- Step 3. Determine the rejection region.
- Step 4. Find the value of the test statistic.
- Step 5. Either reject or fail to reject the null hypothesis.

**Hot Spot #4 Sample Problem and Solution:** Given  $b = 0.3$ ,  $s_b = 0.225$ , and  $n = 20$ , test  $H_o: B = 0$  and  $H_a: B \neq 0$  at the 5% level of significance.

Answer: Step 1.  $H_o: B = 0$  and  $H_a: B \neq 0$ .

Step 2.  $n < 30$  and  $s_b$  is used for  $\sigma_b$ , so we use  $t$ .

Step 3. The critical value of  $t$  for a two-tailed test at  $df = 18$  is  $t = \pm 2.101$ .

Step 4. 
$$t = \frac{b - B}{s_b} = \frac{0.3 - 0}{0.225} = 1.33$$

Step 5. Fail to reject.

**Sample Problem and Answer:** Given  $b = 0.4$ ,  $s_b = 0.173$ , and  $n = 29$  test  $H_o: B = 0$  and  $H_a: B > 0$  for significance at the 5% level.

Answer: Step 1.  $H_o: B = 0$  and  $H_a: B > 0$ .

Step 2.  $n < 30$  and  $s_b$  is used for  $\sigma_b$ , so we use  $t$ .

Step 3. The critical value of  $t$  for a one-tailed test at  $df = 27$  is  $t = 1.703$ .

$$\text{Step 4. } t = \frac{b - B}{s_b} = \frac{0.4 - 0}{0.173} = 2.31$$

Step 5. Reject  $H_0$  in favor of the alternative  $H_a: B > 0$ .

**Sample Problem and Answer:** Given  $b = 0.2$ ,  $s_b = 0.099$ , and  $n = 100$  test  $H_0: B = 0$  and  $H_a: B \neq 0$  for significance at the 5% level.

Answer: Step 1.  $H_0: B = 0$  and  $H_a: B \neq 0$ .

Step 2.  $n > 30$ , so we use the normal distribution for the critical value of  $t$ .

Step 3. The critical value of  $t$  for a two-tailed test at  $df = 98$  is  $t = \pm 1.96$ .

$$\text{Step 4. } t = \frac{b - B}{s_b} = \frac{0.2 - 0}{0.099} = 2.02$$

Step 5. Reject  $H_0$  in favor of the alternative  $H_a: B \neq 0$ .

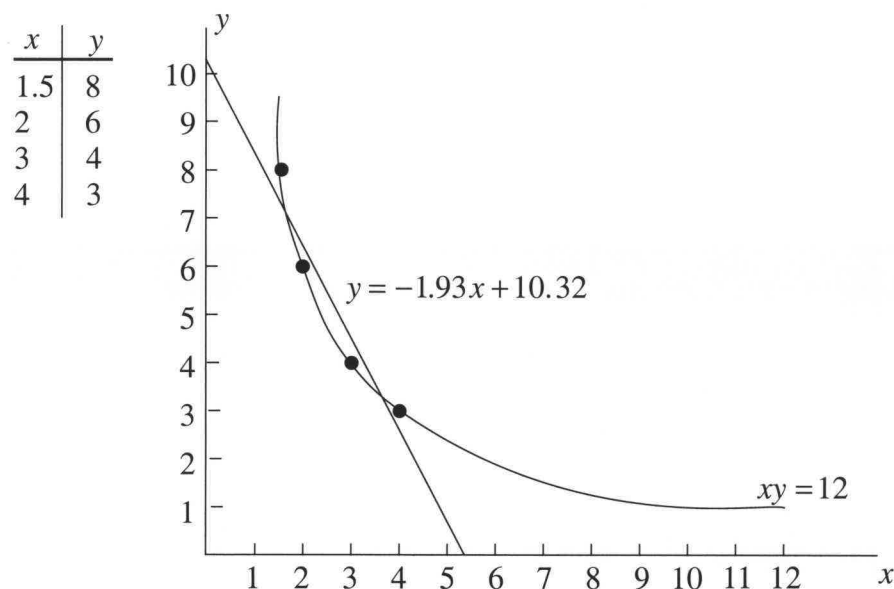
### Hot Spot #5 – The Danger Of Using The Regression Line To Predict Values Of $y$ For $x$ Outside The Interval Used To Obtain The Line

Many times the scatter diagrams appear to be clustered around a line when in fact the relationship between the variables is curvilinear. The apparent linear relationship may be because the values of  $x$  are restricted to a particular interval. The equation  $xy = 12$ , for  $x$  and  $y$  both positive, gives a good illustration of the problems that can exist. The graph of the equation is a hyperbola in the first quadrant as shown below, in Figure 13–9:



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Sections 13.3,  
13.2.5, And 13.9

Figure 12-9:  
A Scatter  
Diagram Of  
Values On The  
Curvi-Linear  
Hyperbola  
 $xy=12$



If the bivariate set of data is limited to values of  $x$  between 1 and 4 as illustrated in the above data set, the graph appears to cluster around the regression line  $y = -1.932x + 10.322$  that is found using the 4 ordered pairs in the table. If we use the regression line to find the value of  $y$  when  $x$  is 10, we get

$$y = -1.932(10) + 10.322 = -9$$

It is clear from the graph of  $xy = 12$  that the  $y$  value must always be positive. The correct value for  $y$  is found by substituting 10 into the equation  $xy = 12$ .

$$y = \frac{12}{10} = 1.2$$

Yet when we use the regression line we get a negative value of  $y$  when  $x = 10$ . The residual  $e$  is

$$e = 1.2 - (-9) = 10.2$$

The error is 850% ( $\frac{10.2}{1.2}$ ). The problem is that the linear relationship we created with the regression line was only valid for the interval from 1 through 4. When we use the regression line to find  $y$  for  $x = 10$  we are assuming that the linear relationship is still valid. In this example we

made a large error in saying  $y = -9$  when  $x = 10$ . As a rule of thumb, we should stay within 20% of the width of our original interval. In the above example the width of the original interval is 3 and 20% of 3 is 0.6.

We can now see how close we are to the correct value of  $y$  when we use the largest value of  $x$  allowed, that is,  $4 + (0.02)(3) = 4.6$ :

$$y = -1.932(4.6) + 10.322 = 1.435$$

The correct value of  $y$  is found by substituting 4.6 into the equation  $xy = 12$ .

$$y = \frac{12}{4.6} = 2.609$$

We still have an error, in that the residual value  $e$  is

$$e = 1.435 - 2.609 = -1.174$$

The error of 45% ( $\frac{1.174}{2.609}$ ) is large; but when compared to the error of 850% that we had using  $x = 10$ , we have a considerable improvement. The point is that we will have an error in either case. But if we limit ourselves to 20% of the width of the original interval for  $x$ , the error will at least be reasonable.

Of course, in practice we never know if the bivariate data set we use for linear regression actually belongs to a curvilinear relation. Relative to the interval (1,4) we used to get the regression line, the correlation coefficient  $r = 0.966$ . When we square  $r$ , we get the coefficient of determination  $r^2$ . This value is 0.933 and it tells us that 93% of the variation in  $y$  is explained by the regression. The point is that relative to our interval of (1,4) the regression line is a good model that explains the clustering we have about the line. It is only when we go outside of the interval that we have the potential for making large errors in estimating  $y$ . Of course, if our correlation coefficient is very small we may still make large errors in estimating  $y$  independent of whether or not there is a curvilinear relationship. Indeed it is possible to have bivariate data sets that exhibit no relationship whatsoever.

## CHAPTER 13 DISCUSSION QUESTIONS

These questions may be used in your study group or simply as topics for individual reflection. Whichever you do, take time to explain verbally each topic to insure your own understanding. Since these questions are intended as topics for discussion, answers to these questions are not provided. If you find that you are not comfortable with either your answers or that your group has difficulty with the topic, take time to meet with your professor to get help.

1. How can you use the graph of the bivariate data to determine if the correlation coefficient is positive or negative?
2. How can you use the scatter diagram to get approximate values of the slope and  $y$  intercept for the regression equation?
3. How is the coefficient of determination used to evaluate the degree of linear relationship between two variables?
4. What is the  $t$  test statistic that is used with hypothesis testing of  $H_o: B = 0$  versus  $H_a: B \neq 0$ ?
5. Why does the interval estimate of  $y$  change as the values of  $x$  are further away from the mean of  $x$ ?
6. Why is a linear relationship between two variables not necessarily a causal relationship?
7. How is it possible that a weak linear relationship between two variables can result in a significant value of  $b$  in a hypothesis test of  $H_o: B = 0$ ?
8. How do errors in the table used for calculations affect both regression and correlation analysis?
9. What are the assumptions of the linear regression model?
10. How might it be possible that two variables are related even though the linear coefficient is nearly zero?